Cheng Chen

Email: 2351445@tongji.edu.cn | chenceason10@gmail.com | ceasonen.github.io | github.com/ceasonen

Education

Tongji University, Shanghai, China Sept. 2023 – Present
B.Eng. in Electrical Engineering, College of Electronic and Information Engineering
Major Rank: **3/87** | CET-6: **519**

Publications

- Yi Xu, **Cheng Chen**, Wenzhuo Lei.
  If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization
  ACM MM 2026 (Accepted; Oral).
- **Cheng Chen**, Yi Xu, Mufan Cao.
  OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment
  IJCAI 2026 (Accepted; Oral).
- Wenzhuo Lei, Mufan Cao, Chaorong Ye, Yi Xu, **Cheng Chen**.
  Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions
  IJCNN 2026 (Accepted).
- Yuyao Cai, Yi Xu, **Cheng Chen**.
  Beyond Relevance: Counter-Evidence Planning in a Controlled Budgeted RAG Testbed
  ICANN 2026 (Accepted).
- **Cheng Chen**, et al.
  Neural-Operator Risk Assessment for Deployable Risk Triage
  NeurIPS 2026 (Under Review).

Research Experiences

If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization
Accepted at **ACM MM 2026**

- Addressed the difficulty of open-vocabulary audio-visual event localization, where heterogeneous audio and visual teachers provide uneven supervision and naive distillation propagates modality-specific errors.
- Proposed **OvOrthKD**, extending orthogonal feature distillation to audio-visual learning so complementary cross-modal cues are preserved while unreliable teacher signals are suppressed.
- Built a lightweight open-vocabulary localization pipeline accepted as an **oral** presentation at ACM Multimedia 2026.

OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment
Accepted at **IJCAI 2026**

- Addressed expertise asymmetry in multi-teacher distillation for diabetic retinopathy screening, where a weaker ViT teacher (**0.83 QWK**) harms students under standard logit fusion with a stronger CNN teacher (**0.88 QWK**).
- Proposed **OrthKD**, distilling only complementary feature-level context from the weak teacher while enforcing orthogonality between teacher projections to avoid redundant knowledge transfer.
- Evaluated on **132,049** retinal images; the **5.4M**-parameter MobileNetV3 student reached **0.885 QWK** on EyePACS and improved zero-shot Messidor-2 performance from **0.507** to **0.728 QWK** (**+22.1** percentage points).

Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions
Accepted at **IJCNN 2026**

- Investigated whether text-only LLMs can perform visual reasoning without multimodal pre-training, avoiding the cost and opacity of end-to-end vision-language models.
- Designed **DeepYOLO**, converting images into grammar-constrained symbolic scene reports with an off-the-shelf detector and feeding them to a text-only LLM.
- On a stratified **5k** VQAv2 subset, achieved **42.3%** accuracy, outperforming a scene-graph baseline (**38.2%**) with **2.5**× fewer prompt tokens; oracle symbolic inputs reached **44.5%**.

Beyond Relevance: Counter-Evidence Planning in a Controlled Budgeted RAG Testbed
Accepted at **ICANN 2026**

- Tackled the limitation of relevance-only retrieval in RAG, where fixed budgets force systems to ignore contradictory evidence and over-trust top-ranked passages.
- Proposed **CapeRAG**, a counter-evidence planning framework that allocates retrieval budget to both supporting and opposing documents before generation.
- Demonstrated more reliable reasoning under strict retrieval budgets by balancing relevance with explicit counter-evidence search rather than greedy top-k retrieval alone.

Neural-Operator Risk Assessment for Deployable Risk Triage
Submitted to **NeurIPS 2026**

- Addressed the deployment risk of neural operators in scientific computing, where incorrect operator predictions can silently corrupt downstream analysis without affordable post-hoc verification.
- Developed a single forward-pass risk assessment module that flags unreliable neural-operator outputs at inference time for deployable triage.
- Targeted low-overhead safety screening integrable into neural-operator pipelines without expensive ensemble or sampling-based uncertainty estimation.

Awards

- National 2nd Prize, Global Campus AI Algorithm Elite Competition (2025)
- National 3rd Prize, Embedded Chip & System Design Competition, FPGA track (2025)

Skills

Programming: Python, PyTorch, Git, Linux, Hugging Face, LaTeX, experiment tracking, LLM/agent prototyping.
Mathematics: Linear Algebra, Probability, Optimization, Signal Processing, Mathematical Modeling.
Language: English — CET-6: 519.